C1.1 Coherency overview

System level coherency enables the sharing of memory by system components without the software requirement to perform software cache maintenance to maintain coherency between caches.

Regions of memory are coherent if writes to the same memory location by two components are observable in the same order by all components.

The ACE protocol enables:

- correctness to be maintained when sharing data across caches
- components with different characteristics to interact
- the maximum reuse of cached data
- a choice between high performance and low power.

The ACE protocol provides a framework for system level coherency. The system designer can determine:

- the ranges of memory that are coherent
- the memory system components that implement the coherency extensions
- the software models that are used to communicate between system components.

C1.1.1 Usage cases

The ACE protocol enables system architects to select the most appropriate technique for sharing data between system components. The protocol does not define specific usage cases, but typical usage cases are:

- the coherent connection of system components
- the coherent connection of subsystems that have non-uniform memory resources
- the coherent connection of components that have a highly optimized local coherency system
- the filtering of coherency communications
- the coherent connection of components that support different coherency protocols, such as MESI, ESI, MEI, and MOESI
- wrapping of components that do not support coherency natively, enabling them to be used effectively within a coherent system level design
- support for cached components that might include multiple levels of cache, and non-cached components
- support for components that store coherency information at different granularities, including cache line granularity and large buffer granularity
- implementations that facilitate optimization of:
  - the primary interconnect within a system
  - multiple subsystems.

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C1.1.2 ACE terminology

Terminology on page A1-25 introduces terminology used throughout the AXI and ACE specifications, and indicates that:

- this specification does not define standard cache terminology, as defined in any reference work on caching
- the *Glossary* defines terms used in the specifications.

ACE introduces additional terms, particularly relating to caching, and to memory operations performed by system masters. The following subsections summarize those terms. Where appropriate, terms listed in this section link to the corresponding glossary definition.

AXI components and topology

The following terms describe components in an AXI4 system. Some terms apply, more specifically, to caches on those components:

- *Caching master*, *Initiating master*, and *Snooped master*
- *Downstream cache*, *Local cache*, *Peer cache*, and *Snooped cache*
- *Main memory* and *Snoop filter*.

Cache state terminology

Cache state model on page C1-135 defines the possible states of a cache entry.

Actions and permissions

The following terms relate to actions that can be performed by a *Master component*, and the permissions to perform such actions:

- *Load*, *Speculative read*, and *Store*
- *Permission to store* and *Permission to update main memory*.

Temporal descriptions

The AXI specification defines *In a timely manner*. The ACE specification requires the additional concept of *At approximately the same time*.

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